CenNano Analysis Toolkit

1. Introduction

CenNano Analysis Toolkit is a desktop application for analyzing nanopore current signals. It integrates two modules:

- Feature Analyzer - loads nanopore traces and extracts statistical and event-based parameters for each signal.
- Signal Divider-Matcher - performs baseline estimation, automatic event detection (signal segmentation), and template matching.

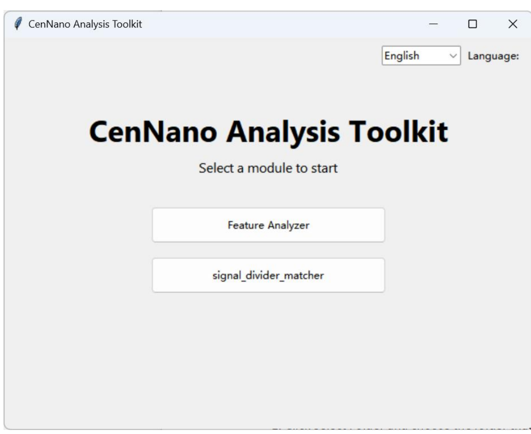
2. Getting Started

1. Double-click CenNano_Analysis_Toolkit.exe to launch the application.

2. The launcher window shows two buttons:

- Feature Analyzer - start the feature-extraction module.
- signal_divider_matcher - start the signal segmentation & matching module.

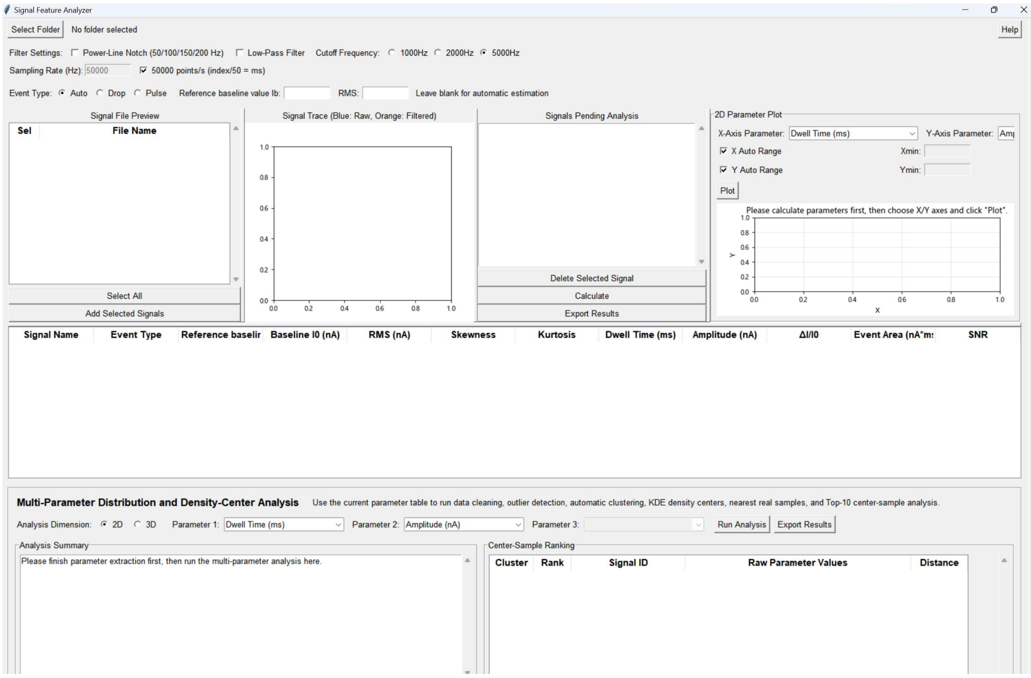
3. Use the Language drop-down in the top-right corner to switch between English and Chinese.



3. Feature Analyzer

3.1 Overview

The Feature Analyzer loads one or more nanopore signals from CSV files, applies optional filtering, and computes a set of statistical and event parameters for each signal. Results are shown in a table and can be plotted as 2D scatter plots or analyzed with multi-parameter clustering.



3.2 Loading Signals

1. Click Select Folder and choose the folder that contains your CSV files.

2. The Signal File Preview list shows all detected files. Click a file to preview its trace (blue = raw, orange = filtered).

3. Use Ctrl/Shift to select multiple files, then click Add Selected Signals to move them into the analysis queue.

4. The Signals Pending Analysis list shows the files to be analyzed.

3.3 Filtering (applied to both plotting and calculation)

- Power-line notch: remove 50/100/150/200 Hz interference.
- Low-pass filter: Butterworth low-pass (4th order by default) with a selectable Cutoff Frequency.

3.4 Sampling Rate and Time Axis

- Current unit: nA; time unit: ms.
- By default, "50000 points/s (index/50 = ms)" is checked, so t(ms) = i/50.
- Uncheck it and enter a sampling rate fs (Hz) to use t(ms) = i x 1000/fs instead.

3.5 Event Type

- Drop - parameters computed using the downward-event definition.
- Pulse - parameters computed using the upward-event definition.
- Auto - computes with both definitions and keeps the type with the larger amplitude.

3.6 Calculating Parameters

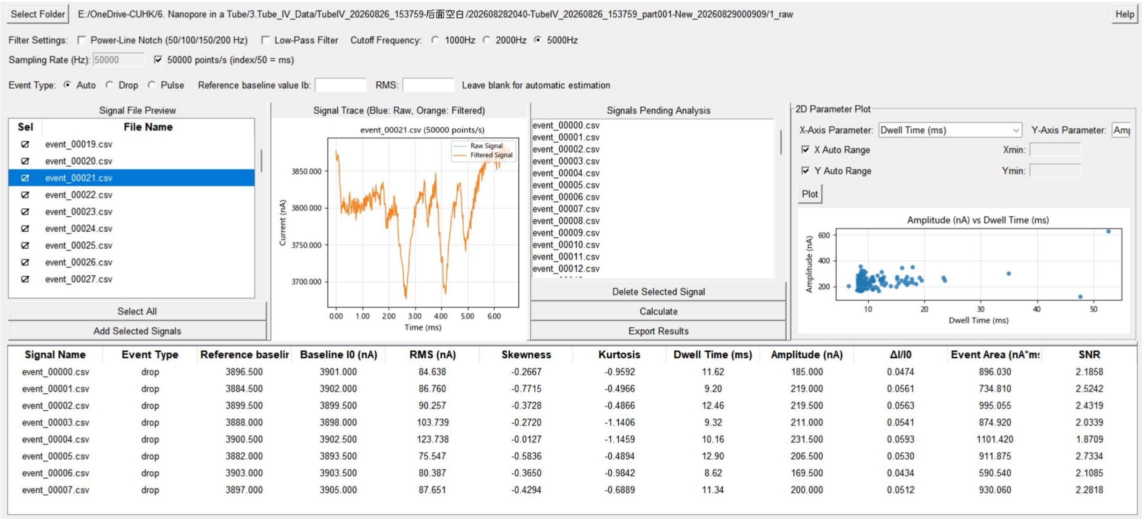
1. (Optional) enter Baseline and RMS values; leave blank for automatic estimation.

2. Click Calculate to generate the result table for all queued signals.

3.7 Output Parameters

The result table contains one row per signal with the following parameters:

Parameter	Description
Signal Name	Identifier of the source file.
Event Type	Drop / Pulse / Auto classification.
Reference baseline value Ib (nA)	Baseline reference; auto = (y[0]+y[-1])/2.
Baseline I0 (nA)	Signal reference value for the event type (I0_drop or I0_pulse).
RMS (nA)	Root-mean-square noise; auto = sqrt(mean((y-Ib)^2)).
Skewness	Skewness of the filtered signal.
Kurtosis	Excess kurtosis (scipy fisher=True).
Dwell Time (ms)	Full-trace duration (max(t) - min(t)).
Amplitude (nA)	Event amplitude on the relevant side.
ΔI/I0	Dimensionless relative current change.
Event Area (nA*ms)	Trapezoidal integral of the event side.
SNR	Signal-to-noise ratio (dimensionless).
Start Time (ms)	Start position of the event.



3.8 2D Parameter Plot

1. After calculation, choose any two columns as X-Axis Parameter and Y-Axis Parameter.

2. Click Plot to generate a scatter plot. Check X/Y Auto Range for automatic axis limits.

3.9 Exporting Results

Click Export Results to save the result table as a CSV file.

3.10 Multi-Parameter Distribution & Density-Center Analysis

This section runs data cleaning, outlier detection, automatic clustering, KDE density-center, nearest real-sample, and Top-10 center-sample analysis on the current parameter table.

1. Choose the Analysis Dimension (2D or 3D).

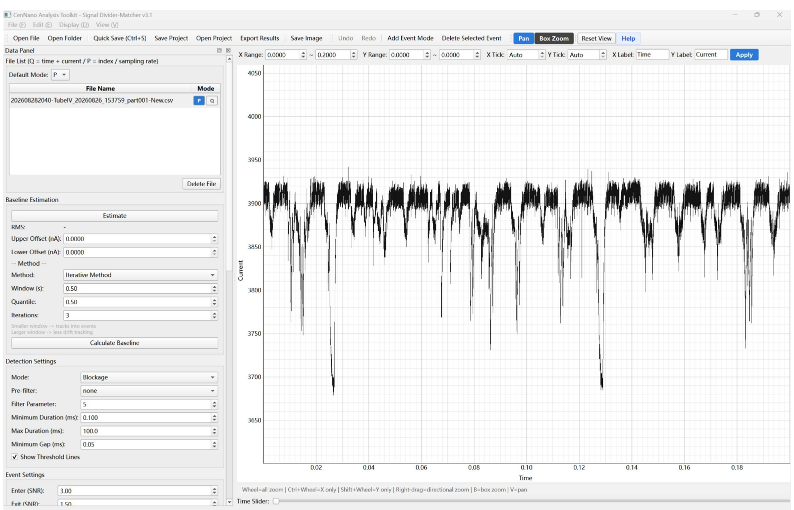
2. Select the parameters (Parameter 1/2/3). Do not select the same parameter twice.

3. Click Run Analysis to view the summary, cluster density centers, center-sample ranking, and analysis plots.

4. Signal Divider-Matcher

4.1 Overview

The Signal Divider-Matcher loads nanopore traces, estimates the baseline, automatically detects events (segments the signal into individual blockade/pulse events), and optionally matches events against templates.



4.2 File Reading Mode

- Set the default mode P/Q at the top of the file list; newly imported files use it by default P mode.
- Each file row has its own P/Q toggle to change the mode individually.
- Q-Mode: the CSV already contains time and current columns.
- P-Mode: the CSV stores current by index (no time column); time = index / sampling rate (default 50,000 Hz).

4.3 Baseline Estimation and RMS

- Estimate (drag to calculate RMS): drag over an event-free baseline segment on the plot to compute the RMS (used for SNR threshold conversion).
- Upper/Lower Offset (nA): offset the baseline by +/- to form the baseline region between two boundary lines.
- Baseline estimation methods: sliding quantile / morphological opening / iterative method.

4.4 Event Detection Settings

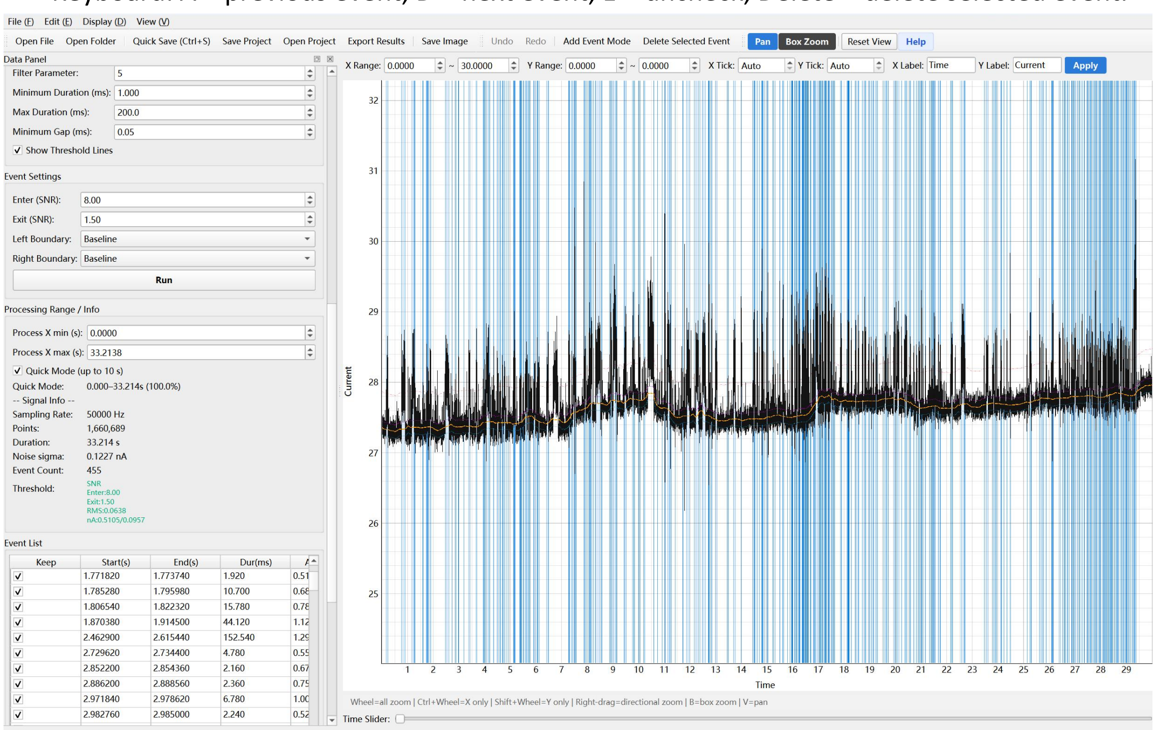
- Blockade mode: detects downward deviations (dev = baseline - signal).
- Pulse mode: detects upward deviations (dev = signal - baseline).
- SNR threshold: enter_th = SNR_enter x RMS; exit_th = SNR_exit x RMS.
- Left/Right boundary: baseline / baseline-region upper bound / baseline-region lower bound.

4.5 Processing Range and Quick Mode

- Process X min/max: restrict analysis to a specific time segment.
- Quick mode: limits detection to at most 10 seconds.

4.6 Event List and Shortcuts

- The leftmost Keep checkbox is checked by default; unchecked events are not exported.
- Keyboard: A = previous event, D = next event, E = uncheck, Delete = delete selected event.

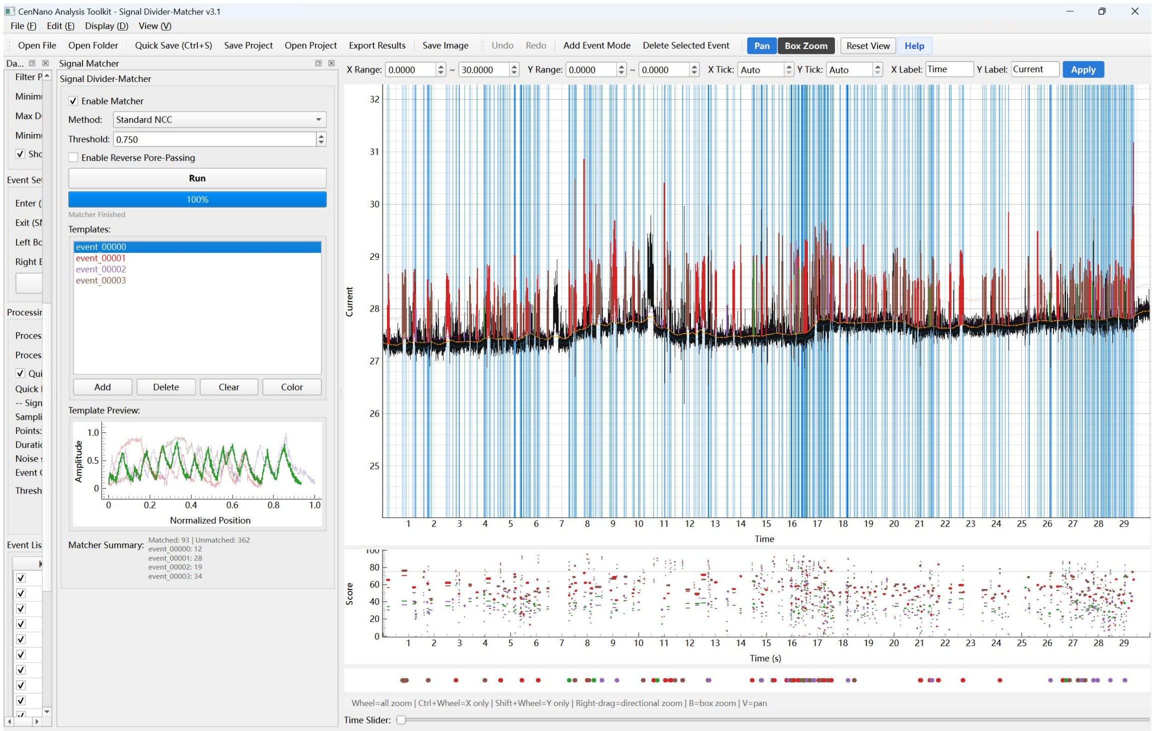


4.7 Template Matching (Matcher)

1. Enable Matcher panel in View, then load templates via Add Template.

2. Run matching; each event is scored against the templates (NCC + shape difference).

3. The score plot for each template is shown below the main signal plot.



4.8 Export

- Only Keep-checked events are exported.
- A folder <original name>_YYYYMMDDHHMMSS is created to avoid overwriting.
- With Matcher enabled, subfolders all_events/, matched/, unmatched/, and by_template/ are also exported.

5. Appendix

5.1 Keyboard Shortcuts (Signal Divider-Matcher)

Key	Action
A	Previous event
D	Next event
E	Uncheck the selected event (keep in list)
Delete	Delete the selected event
B	Box zoom
V	Pan mode
Mouse wheel	Zoom all axes
Ctrl + wheel	Zoom X axis only
Shift + wheel	Zoom Y axis only
Right-drag	Directional zoom

5.2 Export Folder Structure (Signal Divider-Matcher)

```
1_raw/         Raw segmented signals

2_blockade/    Blockade mode (downward)

3_pulse/       Pulse mode (upward)

output.json    Parameters and event info

all_events/    All events (when Matcher enabled)

matched/       Matched events

unmatched/     Unmatched events

by_template/   Grouped by template
```